

Leveraging Transdisciplinary Engineering Through the Coalescence of Digital Twins and XR-Technologies

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Abstract. Digital Twins (DTs) as a disruptive technology of digital transformation are emerging as a true game changer in several application fields. Through digital representations of physical entities, vertical and horizontal integration can take place. This is the basis for comprehensive interoperability and accessibility between human and machine. This paper elaborates an application scenario bringing together DT in a collaborative environment with manifold devices like computers, tablets, or Extended Reality (XR) devices. Therefore, a DT has been developed which can describe and adapt itself according to the user requirements. This is complemented by a cross-compatible application that visualizes the sensor data in almost real-time and allows the user to interact with or control the DT from any location. In a first proof of concept, a seminar room has been equipped with interior air quality measurement devices which encapsulate several sensors within a 3D printed surrounding. The resulting data was then made available via the corresponding DTs of the measurement devices to the user on a HoloLens 2, Varjo XR-3, computer, or mobile device. The bidirectional flow of data and information between the physical and digital space enables to influence the physical devices through their DTs. The DT, combined with XR, can be used transdisciplinary in teaching or industrial environments.

Keywords. digital twin, virtual reality, augmented reality, extended reality, advanced user interface, transdisciplinary collaboration

Introduction

Due to their interdisciplinary character and domain independence, Digital Twins (DTs) are a key technology of both Industry 4.0 implementations and Internet of Things (IoT) systems [1]. Through controversial discourse, transdisciplinary approaches can thus also be realized. Extended Reality (XR) and related technologies perfectly complement the premises of DT-systems for monitoring, operation, and feedback purposes to the physical world [2]. Therefore, a significant trend of XR and DT as a fusion of physical and virtual space can be observed in future industry [3].

To bridge this gap a bit further, this paper outlines a way of realizing an environment for interdisciplinary collaboration of DTs in coalescence with XR technologies. For this purpose, an architecture is implemented based on identified requirements and its

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feasibility is proven within a functional proof of concept. As a demonstration object, for the sake of comprehensibility, a methodology for assessing indoor air quality is employed. Indoor air quality in educational facilities is usually inadequate due to improper ventilation practices. The creativity and ability to focus of both students and teachers directly depends on the air quality. An improvement of the carbon dioxide concentration (CO_2) causes a proven average increase in performance [4]. The developed air quality measuring devices are placed in the seminar rooms and autonomously provide their DTs within the infrastructure. A cross-compatible application as a powerful back-end of the DT layer allows data access to the physical devices in near real-time and can reflect actions and instructions back to the devices in a bidirectional way. The DTs can thus be visualized and influenced from state-of-the-art XR devices, such as *HoloLens 2*, *Varjo XR-3*, computers, and mobile devices which are spatiotemporally independent, hence establishing an interdisciplinary collaboration scenario which could easily be transferred to a variety of engineering disciplines.

This paper contributes: (1) an approach to the coalescence of DTs and an immersive XR environment for interdisciplinary collaboration across disciplines facilitating transdisciplinary working modalities; (2) an implementation strategy for the representation, management, and interconnection of data-driven DTs; (3) the scaling of different XR applications to cross-compatible end devices.

First, the related work is outlined. On the one hand, DTs are focused while on the other hand XR technologies are highlighted in section 1. After deriving requirements for interdisciplinary collaborative environments therefrom in section 2, section 3 describes the application use case in order to monitor indoor air quality. Section 4 is dedicated to the implementation and is finalized by a functional proof of concept and the evaluation in section 5. The discussion of the realized structure is carried out in section 6 and finalized by a brief conclusion with prospects in section 7.

1. Related Work

DTs are an enabler technology paving the way for the future industrial revolution [5]. As a disruptive concept, DTs are key to facilitate digital transformation and can be applied in many domains, regardless of their use case [1]. The concept was established in 2002 by Michael Grieves as an optimum for product life cycle management. The duality of his approach is divided into physical and digital space. The physical space contains the real entities, termed as Physical Twins (PTs), which are mirrored in the digital space as a 1:1 representation with sufficient application-related accuracy. The bilateral flow of data and information between the two spaces is referred to as the digital thread [6, 7]. DTs can significantly simplify the effort required in the engineering as well as the operating phase of a device [8].

By enriching the concept with other key competences of digital transformation, such as Virtual Reality (VR), Augmented Reality (AR), Mixed Reality (MR), far greater value can be generated [2]. Lehmann et al. [9] show how DTs can be empowered to collaborate with others and thus interact and negotiate interdisciplinary across company boundaries. Transdisciplinary collaborations enabled by DTs in different virtual environments can be helpful in many engineering scenarios [10]. DTs also enable a suitable interface to humans that satisfies future requirements in terms of human-centeredness of technology [11].

The different environments can be divided into three technologies which distinguish in their goal of manipulating humans and their stimuli. VR attempts to address as many human senses as possible with synthetically generated signals. This immersion is the biggest difference to other Human-Machine-Interfaces (HMIs) or 3D programs such as CAD software which represent 3D objects on a 2D medium using computer graphics [12]. In contrast, AR combines virtual and real impressions, and the real world is overlaid with virtual objects – these holograms will act as if they were real. MR is situated in the middle of the Reality-Virtual continuum and allows virtual and real content to be combined in such a way that the user can no longer distinguish from reality [13].

These three categories are grouped under the umbrella term XR, which has become more importance due to the necessary social distancing caused by the COVID-19 pandemic. This makes it possible for experts from different departments to collaborate transdisciplinary over long distances or even simulations could be graphically processed [14, 15, 16]. But also in teaching or training scenarios from different areas, this technology gets more relevance, because it allows to represent complex subjects in space or different interventions can be repeated several times [17, 18, 19, 20]. If this visualization is combined with intelligent interaction and the DTs, it results in broad information transfer for everyone and each field.

2. Requirements on Interdisciplinary Collaboration Environments for Digital Twins in Coalescence with XR Technology

According to Onaji et al. [21] and Lober et al. [22], specific requirements on DT environments can be determined as follows: **Replication and fidelity** must be as accurate as necessary and as simple as possible, tailored to the use case. The **identity** between the twins must be clearly identifiable. The **connectivity, data aggregation and storage** must support common IoT interfaces and, in addition to storing the data, must also fulfill the bidirectional interlinking of the twins. The **latencies and computational power** must be coordinated with the use case and must be sufficiently provided to the DT. The **integration with back-end applications** through superimposed intelligent features must be ensured. **Security, privacy, and integrity** must be compliant with modern IT standards.

Further, for an interdisciplinary, collaborative environment, the amalgamation with DT environments and XR environmental constraints must also be considered. Enabling a live connection between the physical resources and a virtual application layer, a **powerful back-end** for virtual technologies is required. This architecture should allow the **integration of various hardware** components in a dynamic and flexible way. In order to achieve this **real-time dual-direction data communication** is necessary to monitor incoming data in the virtual environment and simultaneously transmit virtual interactions to the physical device. A **game engine** is therefore needed, which is also established in the game and movie industry. But not only here, as gamification is increasingly finding its way into other areas such as engineering, construction, or education, as shown in Beisheim et al. [10] and Koca et al. [23].

3. Application Use Case Description

Following the identification of related work and the derivation of corresponding requirements, the presentation of the architecture can be supplemented by an illustrative use case. The measurement of CO₂ concentration is an adequate medium for monitoring indoor air quality [4]. Thermal comfort in terms of temperature and humidity is also considered as a key factor in well-being and productivity [24]. Especially in the context of the global COVID-19 pandemic, the acquisition of such quality parameters has come into focus.

For monitoring the teaching and laboratory facilities, microcontroller-based measuring devices were developed. This will provide the basis for collaborative, interdisciplinary monitoring, and response to air quality issues in seminar rooms, which constantly arrive from changing student groups at varying time intervals. The basis for realization is an *ESP8266* microcontroller from *Espressif Systems*, which is connected to two sensors for detection of air quality parameters. In addition to the cost-effectiveness, number of General Purpose Inputs/Outputs (GPIOs) as well as interfaces, it offers the great advantage of a Wi-Fi interface for unrestricted connection to a wide range of IoT systems. The sensors chosen were the *MH-19ZC NDIR* CO₂ sensor from *Winsen Electronics* and the *BME280* compact sensor from *Bosch Sensortec* capturing relative humidity, barometric pressure, and ambient temperature, as both are market leaders in terms of price and quality. Complemented by a 4 row LCD display, everything is housed within a self-designed and 3D printed robust enclosure.

4. Architecture and Implementation of an Interdisciplinary Collaboration Environment for Digital Twins in Coalescence with XR Technology

With the described use case as a foundation, the architecture and its implementation can now be elaborated. The architecture of the interdisciplinary collaboration environment is divided into three main areas as depicted in Figure 1. According to Grieves' concept of DTs, all physical entities in the form of the air quality measurement devices are accommodated in the *Physical Twin Layer*. The DTs as virtual representations are located in the overlaying *Digital Twin Layer* including management and higher-level handling functionality. As further superordinate applications, XR appliances serve for interaction and visualization of the DTs in order to create a foundation for collaboration in order to create a transdisciplinary basis for discussion.

In more detail, the microcontroller-based physical devices in the *Physical Twin Layer* are programmed in C++ and have the ability to provide themselves as DT in the parent layer in terms of their structural as well as semantic description. For this purpose, Hypertext Transfer Protocol (HTTP) messages are sent to the relevant software components for registration and certification purposes. The subsequent telemetry transmission of the sensors then takes place under the exchanged credentials via Message Queuing Telemetry Transport (MQTT).

For the realization of the *Digital Twin Layer* the combination of the two open-source *Eclipse* projects *Hono* and *Ditto* are utilized. As a universal middleware, *Eclipse Hono* can interconnect a wide range of different IoT devices with back-end applications in a modular and scalable way. A major advantage here is the bidirectional normalization of the data, so that protocol independence is achieved. In addition, device clusters are

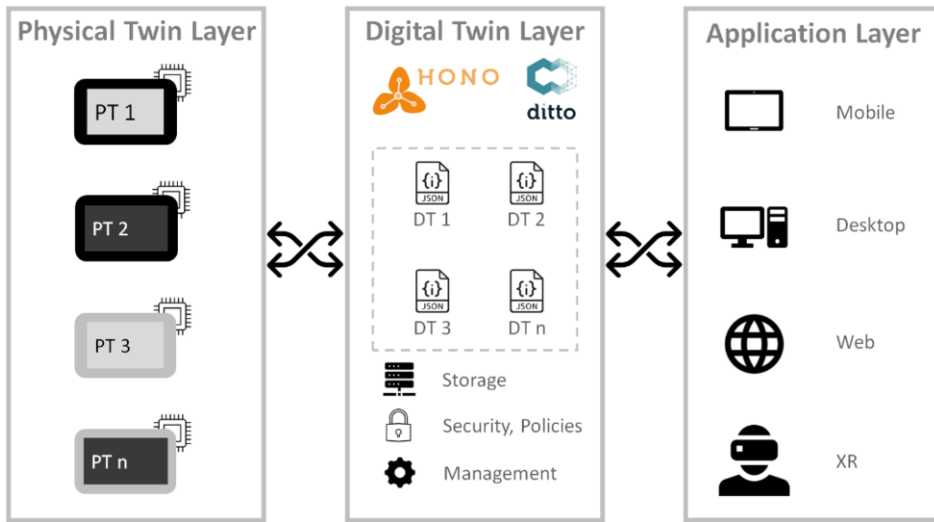


Figure 1. Architectural Implementation Concept of the Interdisciplinary Collaboration Environment for DTs in Coalescence with XR Technologies.

established in the form of tenants, and additionally state-of-the-art security mechanisms are implemented. The deployment process in *Eclipse Hono* originated from the PT is divided into the creation of the tenant, the creation of the device and the declaration of its credentials. *Eclipse Ditto* as a universal DT provider is directly linked to *Eclipse Hono* and is responsible for the structure and data representation of the PTs in the form of a dynamic JavaScript Object Notation (JSON). This abstraction layer, with its universal bidirectional interfaces and extended policy handling, enables the easiest possible access to DTs. For example, agile forwarding of modifying DT data with specific payload mapping can also be defined here. In order for the PTs to provide themselves here, they must also set a policy, create the device, and declare the device features.

In the final step, applications at the *Application Layer* can access DTs at the *Digital Twin Layer* from the back-end and thus also reflect commands back into the physical world. Based on the requirements, the core of the *Application Layer* is the *Unity* development environment, because the support and community are much higher than on other engines like Unreal for example. The *Unity* framework is a cross-platform game engine with the opportunity to develop 2D and 3D games [25, 26]. In the beginning, a 3D core project was created, and the *OpenXR* Application Programming Interface (API) was integrated. The open-source standard allows the created scene to port it to multiple virtual and augmented reality devices such as *HoloLens 2* or *Varjo XR-3* without creating a separate XR project for each device. This saves time and makes it easier to develop an immersive environment without spending a lot of time on the technical details of the prototyping process.

After the application is connected to a network, the client for the machine-to-machine communication tries to authorize a connection with a predefined MQTT server (broker) which is fed by the previously mapped and forwarded DT data from *Eclipse Ditto*. Once the connection is successfully established the client subscribes with

predefined topics with an associated Quality of Service (QoS). To keep data traffic as low as possible and focus on real-time data, a QoS of 0 has been chosen. However, the user can adjust the broker configuration and connection parameters at runtime if the requirements on the communication partner or other specification will change. This flexibility makes it possible to dynamically adapt the MQTT configuration to the current conditions.

When the application receives telemetry data from a physical device via the *Digital Twin Layer*, it first checks the twin status based on the unique device ID. If the device is already virtually created at the *Application Layer*, the parameters will be updated. However, if it is a new or unknown DT, a virtual representation of the device is created in the background of the *Application Layer*. Via the User Interface (UI), the user sees the current status of the DT as well as the number of available DTs in the current contextual situation. To avoid overloading the UI, only one visual representation of the DT is visible. The interaction buttons enable to control or switch between the different twins. When switching, the last known telemetry data of the selected twin is displayed. In addition, the visual representation has extra features that the physical device does not have. Besides supplementary measurement values and timestamps, the XR environment displays meta information as well as a button for switching the device back-light on and off. The reflection on the physical device demonstrates the bidirectionality of a DT as required by definition.

5. Proof of Concept & Evaluation

After the successful implementation of the presented architecture, a functional proof of concept will be carried out and subsequently evaluated. Figure 2 shows the feasibility of a dynamic cyber-physical system that was successfully established enabling the coalescence of DTs and XR technology.

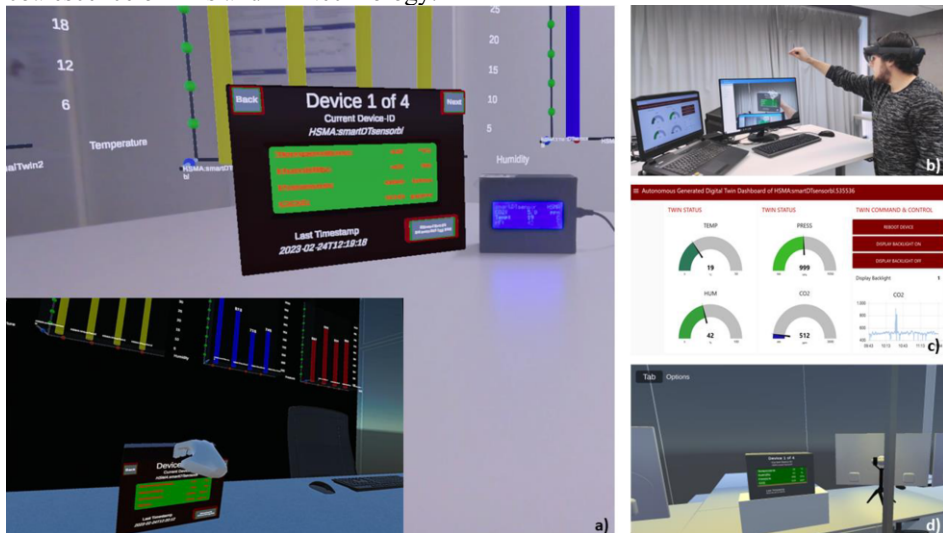


Figure 2. Collaborative Multi-User Session between the Universities of Applied Sciences Alb.-Sig. and Ma. Different Device Views of the DT Visualized in the Application Layer (a) Varjo XR-3 View in Video Pass-through and in the Lower Left Corner the VR View (b) AR Representation with the HoloLens 2 (c) Dashboard View via Web Browser (d) 2D View on the Computer or Tablet (a,d) Ma. Users (b,c) Alb.-Sig. Users.

On the Application Level, the user can utilize different devices such as PC, mobile devices, or advanced Head-Mounted Displays (HMDs). Therefore, these users are not limited to a specific system or setup and can choose the entry point into the XR-technology that fits best. From the technology view, the use of Unity has made it possible to connect state of the art HMDs such as the HoloLens 2 or Varjo XR-3 on the one hand, but on the other hand systems such as PC or mobile devices can also join the session without big additional effort. This ecosystem allows users from different departments to communicate, collaborate with each other and control their systems even if they are physically separated, as shown in Figure 2. The University of Applied Sciences Albstadt-Sigmaringen is represented in the session with a HoloLens 2 (see Figure 2b) and via a web browser (see Figure 2c) and the University of Applied Sciences Mannheim join the session with a PC and the Varjo XR-3 (see Figures 2a and 2d). Based on the plug-and-play realization on the hardware side, the air quality measurement devices only need to be connected to power and the current air quality from the differently positioned sensor boxes in seminar rooms is visualized. Every end user is able to interoperate with the individual devices via the corresponding interaction interface. Through this, for instance, appeals and instructions for better ventilation behavior can be enforced in the seminar rooms without the need for physical presence.

In addition, a first qualitative survey of users and experts from various fields such as biotechnology, mathematics, engineering, and security showed a positive feedback on the combination of DT and XR systems compared to 2D representations. This was explained by the intuitive control via hand-tracking and not as normal via the traditional 2D input devices such as mouse and keyboard. Because there is the possibility to directly interact with the 3D subject without having an immersion break. Another positive remark was the indicator of the AR data glasses. This feature helps the persons to better orient themselves in the AR environment, as it appears as soon as the virtual object is out of the field of view and makes it easier to find the virtual object again. A further positive feature is the opportunity to switch between VR and AR with the Varjo XR-3. This gives the users the option of including their environment or separating from the real world if necessary. This seamless integration of physical and digital content allows for an even more immersive experience than an environment is exclusively 2D visualized. While 2D interfaces still have their place, XR systems open up new, creative ways for users to interact intelligently with things and are therefore an excellent complement. All these characteristics reinforce the architecture with the efficient transdisciplinary engineering scenario.

6. Discussion

This paper describes an approach of how transdisciplinary engineering can be leveraged by the coalescence of interdisciplinary DTs and XR technology. Based on different from the literature derived requirements on DTs and XR as well as a demonstrative use case, an architecture and its implementation synthesizing both technological domains could be established.

Regarding the requirements, sufficient **replication and fidelity** could be achieved by employing *Eclipse Ditto*. The JSON-based representations offer sufficient freedom to describe the structure and semantics of the PTs. This data format allows an individual representation of all structures from every field, a technical-level exchange can be done and it is readable for everyone. **Identity** is satisfied via unique tenants and IDs within

both *Eclipse* projects. For an adequate **connectivity, data aggregation and storage** *Eclipse Hono* is serving with its various IoT interfaces. Aggregation and storage of the data is managed within *Eclipse Ditto*. By deploying in a scalable microservice-based architecture, **latencies and computational power** can also be appropriately mapped. The entire implementation and the proof of concept have shown that a **back-end integration** with superimposed intelligent features is also easily possible independent of the people and their field of expertise. **Security, privacy, and integrity** is also provided by both *Eclipse* projects so that a proper DT management could be build. With *Unity*, a universal **powerful back-end** implementation framework for all XR applications was available. Regardless of the **various XR hardware** used, it was possible to implement a **real-time dual-direction data communication**.

Especially after the elaboration of the requirements derived from the related work, it becomes apparent that DT technologies and XR approaches complement each other very well and can mutually reinforce each other as enabler technologies.

However, it should be noted that DTs are not limited to the visual representation within XR environments. Depending on their use-case, DTs play out their true strengths in simulation, data integration, intelligent collaboration with each other etc. It should also be mentioned that the development of the air quality measurement devices and their use-case were not an essential part of the work and were only used for transdisciplinary illustrative purposes to achieve an air ventilation strategy for optimal indoor air quality for a better learning environment and to maintain the performance and creativity of students and teachers. In addition, this strategy also allows remote users who are not in the room to notify on-site users as soon as a change in measurement results is detected.

The presented architecture pays back its true strengths in the case of arbitrarily scalable, domain-independent objects. For example, users from different disciplines (electrical engineering, mechanical engineering, informatics, mathematics etc.) could gain different insights and contexts on the DT of an object through transdisciplinary collaboration.

In the future, these XR collaboration scenarios could be improved by incorporating other tracking methods (e.g., eye-tracking). The integration of avatars of partaking users could further enrich the user experience. Voice commands between the different participants or bio signals through brain-computer interfaces could also be interesting research appliances to bring academic people, medical scientists, interested industry professionals and engineers even closer together.

7. Conclusion & Future Work

To leverage the experience of transdisciplinary engineering the coalescence of DT and immersive XR technology is placatively illustrated. Therefore, an architecture is implemented based on identified requirements and its feasibility is proven within a functional proof of concept. To substantiate this approach a comprehensible use case in the form of air quality monitoring is utilized.

In the prototype a bidirectional connection has been successfully established which allows to visualize live data in the digital space and in the same point to control the DT independently of the user's discipline. The DTs can thus be visualized and influenced from state-of-the-art XR devices, such as *HoloLens 2*, *Varjo XR-3* or computers and mobile devices, independent of location and time establishing a transdisciplinary collaboration scenario which could easily be transferred to a variety of engineering

domains (electrical engineering, mechanical engineering, informatics, mathematics etc.). This realistic scenarios with real-time data separately of the field and offers the opportunity to train user or explain complex issues to them. It becomes apparent that DT technologies and XR approaches complement each other very well and can mutually reinforce each other as enabler technologies.

This paper contributes: **(1)** an approach to the coalescence of DTs and an immersive XR environment for interdisciplinary collaboration across disciplines facilitating transdisciplinary working modalities; **(2)** an implementation strategy for the representation, management, and interconnection of data-driven DTs; **(3)** the scaling of different XR applications to cross-compatible end devices.

However, the use case shown is very limited and could be scaled to any complexity in the future. Likewise, DTs are not tied to visual representation, but can exhibit their strengths in manifold applications through their harmonized data structure. In further studies the architecture will be adapted to other use cases and the interaction design will be made more intuitive through the utilization of other technologies (e.g., different tracking methodologies, voice commands, brain-computer interfaces). Furthermore, it is planned to integrate avatars, which are represents of the individual users and bringing together academic people, medical scientists, interested industry professionals and engineers.

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